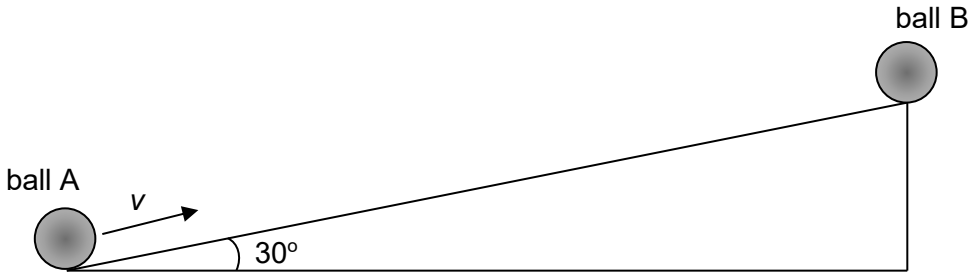
2 Ball A is launched up the slope from the bottom of a smooth inclined plane with an initial speed v. At the same time that ball A is launched, ball B is released from rest at the top of the inclined plane. The two balls meet at the mid-point of the inclined plane after two seconds.



Given that the mass of ball A is twice that of ball B, what is the speed v?

A 4.9 m s^{-1}

B 9.8 m s^{-1}

C 17 m s^{-1}

D 20 m s^{-1}